OFFICIAL TECHNICAL SPECIFICATION

SportSense AI Pro

Complete Architecture, Edge Computer Vision & Biometric Proctoring Guide

A comprehensive engineering reference documenting the end-to-end architecture, on-device AI kinematics, two-phase facial biometric anti-malpractice lifecycle, talent identification formulas, and full system implementation.

COMPUTER VISION ENGINE

MediaPipe Tasks-Vision

60 FPS on-device 33-point 3D pose tracking via WebGL & WASM

BIOMETRIC INTEGRITY

Two-Phase Proctoring

Face ID enrollment on Test 1; Anti-proxy & multi-person defense on Tests 2-5

AUDIO DEFENSE SYSTEM

Web Audio API Siren

Zero-asset browser synthesized 640-960 Hz security alarm sweep

CORE ATHLETIC TRAITS

5 Olympic Vectors

Explosive Power, Agility, Balance, Strength & Coordination

Platform: React 19 • TypeScript 5.8 • TanStack Start • Supabase **Author:** SportSense AI Engineering Team • 2026

1. Project Overview & Core Mission

SportSense AI Pro is an edge-AI athletic talent assessment and biomechanical movement analysis platform. It empowers athletes, coaches, and sports organizations to conduct standardized physical tests using consumer smartphones or webcams, extracts real-time kinematic measurements, safeguards battery integrity through biometric anti-proxy proctoring, and recommends optimal athletic sports pathways.

Core Breakthroughs

- **Zero Expensive Lab Hardware:** 100% in-browser edge computing with zero cloud video upload latency.
- **Biometric Test Integrity:** Eliminates substitute/proxy athletes across multi-test assessment batteries.
- **Scientific Talent Discovery:** Normalizes biomechanical data into Olympic-level talent suitability scores.

2. Complete Technology Stack

Layer	Technologies	Key Responsibilities
Frontend Core	React 19, TypeScript 5.8	Type-safe component architecture, strict React 19 hooks, UI lifecycle
Routing & SSR	TanStack Start, TanStack Router 1.170, Nitro	File-based type-safe routing, SSR hydration, route parameters
Styling & Design	Tailwind CSS v4, Radix UI Primitives, Lucide	Accessible primitives (dialogs, drawers, tabs), responsive dark/light modes
Edge Computer Vision	MediaPipe Tasks-Vision 1.0 (PoseLandmarker)	WASM & GPU-accelerated 33-point 3D pose and multi-person tracking
Biometric Proctoring	IPD Geometry Vector Engine, Web Audio API	Scale-invariant facial ratios, synthesized dual-tone siren alarm sweep
Backend & Database	Supabase (PostgreSQL, GoTrue Auth, RLS)	Athlete authentication, profile sync, assessment run logs, secure storage
Data Visualization	Recharts 2.15, HTML5 Canvas API	Real-time radar charts, trait suitability bars, printable PDF certificate
Build & Tooling	Vite 8, ESLint 9, Prettier 3	Sub-second HMR dev server, optimized rollup bundling, code quality

3. High-Level System Architecture

1 **Authentication & Profile Calibration**

Athletes authenticate via Email, Phone OTP, or Google OAuth. Profile inputs (age, height, fitness level, preferred sports) compute a 0-100% calibration score before testing unlocks.

2

Edge Vision & Kinematic Analysis

MediaPipe tracks 33 3D skeletal nodes at 60 FPS. Trigonometric vector algorithms compute joint flexion angles (e.g. knee depth) and center-of-mass trajectories in real time.

3

Two-Phase Biometric Proctoring Defense

Test #1 captures facial geometry and locks the athlete's Face ID. Tests #2-5 continuously monitor for unauthorized proxies or multiple persons, triggering a siren and 30s abort timer on violations.

4

Talent Discovery & Sport Recommendation

Synthesizes the 5-test battery into 5 foundational traits (Explosive Power, Agility, Balance, Strength, Coordination) to rank suitability for Volleyball, Basketball, Football, Cricket, Badminton, and Tennis.

4. Computer Vision & Edge Kinematics Engine

The vision pipeline operates entirely in-browser using **Google MediaPipe Tasks-Vision**, executing inside WebAssembly with WebGL GPU delegation to achieve 60 FPS without server streaming overhead.

33-Point 3D Pose Mesh

Tracks anatomical nodes across the head, shoulders, elbows, wrists, hips, knees, ankles, and feet. Each node reports normalized coordinates `(x, y, z)` and presence confidence scores.

Multi-Pose Detection (numPoses: 3)

Configured to detect up to 3 persons concurrently. Filters out noise by requiring at least 5 nodes with visibility ≥ 0.35. Highlights secondary persons in red/amber overlays.

Joint Angle Calculation Trigonometry

Flexion and extension angles (such as knee bend during squats or landing cushion during jumps) are calculated using 2D/3D planar vectors:

```
Angle = | atan2(y3 - y2, x3 - x2) - atan2(y1 - y2, x1 - x2) | * (180 / π) // Normalized to 0° - 180° range (Hip = p1, Knee = p2, Ankle = p3)
```

5. Two-Phase Biometric Identity & Anti-Proxy Proctoring

To guarantee test authenticity, SportSense AI Pro implements an active two-phase biometric proctoring lifecycle that distinguishes between authorized athletes, unauthorized proxies, multiple persons, and temporary room departures.

PHASE 1: Assessment #1 (Enrollment)

- **Zero Malpractice Triggers:** Checks are completely disabled during Test 1.
- **Empty-Frame Immunity:** An empty room will NEVER lock an empty profile.
- **6-Frame Stability Gate:** Once exactly 1 athlete is visible for 6+ stable frames, the system extracts the IPD vector and captures a 128x128 face thumbnail.
- **Auto-Profile Sync:** The photo is immediately synced to `/profile` with a "Biometric Face ID Enrolled" badge.

PHASE 2: Assessments #2–5 (Active Defense)

- **Empty-Room Immunity:** Stepping away or adjusting equipment triggers NO alarms. The HUD displays "Waiting for athlete...".
- **Multi-Person Tamper Defense:** >1 person sustained for 2.0s triggers malpractice.
- **2-Second Analysis Window:** Evaluates facial geometry over a 2s buffer before triggering alarms, eliminating motion-blur false positives.
- **Full Lockdown:** Side-by-side face modal, synthesized siren alarm, and 30s auto-abort timer.

Inter-Pupillary Distance (IPD) Facial Biometrics

Facial landmarks are normalized by the Inter-Pupillary Distance (distance between pupils) to ensure absolute scale and distance invariance:

```
IPD = sqrt((x_leftEye - x_rightEye)^2 + (y_leftEye - y_rightEye)^2 + (z_leftEye - z_rightEye)^2) Biometric Ratios = [ Eye-to-Nose / IPD, Mouth-Width / IPD, Nose-to-Mouth / IPD, Face-Breadth / IPD, Vertical-Height / IPD ] Weighted Similarity = 0.25*S_mouth +
```

```
0.20*S_eyeNose + 0.20*S_noseMouth + 0.20*S_breadth + 0.15*S_height // Threshold: Match >= 76% (Authorized); Mismatch < 76% (Proxy detected)
```

> 🚨 **Malpractice Reaction & Recovery Protocol**
>
> When an unauthorized proxy or multiple persons persist past 2 seconds:
>
> 1. **Side-by-Side Face Display:** Renders registered athlete photo vs detected proxy snapshot.
> 2. **Continuous Web Audio Siren:** Synthesizes a loud 640 Hz to 960 Hz dual-tone police alarm.
> 3. **Hard Controls Lock:** Video recording and submit actions are disabled.
> 4. **30s Recovery Countdown:** If the original athlete returns within 30s, the alarm clears and recording unlocks. If 30s expire, the session aborts and redirects to `/dashboard`.

6. The Standardized 5-Assessment Testing Battery

The assessment battery consists of five standardized exercises that collectively test the athlete's kinetic chain, neuromuscular coordination, and athletic foundation:

Test	Primary Metric	Biomechanical Objective	Proctoring Role
1. Vertical Jump	Jump Height (cm)	Flight time, explosive displacement, rate of force development (RFD)	Enrolls athlete face geometry and profile photo
2. Squat	Depth Score (80°-90° target)	Eccentric knee/hip control, valgus alignment, movement tempo	Active verification against enrolled face
3. Single-Leg Balance	Stability Index (0-100)	Unilateral proprioception, ankle stabilizer recruitment, CoM sway	Continuous single-person stability monitoring
4. Push-Up	Valid Reps in 60s	Upper-body muscular endurance, scapular stability, full lockout	Side-profile facial continuity check
5. Side-to-Side Agility	Contacts / 20s	Lateral change-of-direction speed, deceleration & reactive force	Guarantees registered athlete performs high-speed drill

7. Talent Identification & Sport Suitability Algorithm

Raw performance metrics are normalized into five core athletic traits on a 0–100 scale:

1. Explosive Power

Vertical Jump height & rapid hip-knee extension velocity.

2. Agility

Lateral change-of-direction contact frequency and footwork tempo.

3. Balance

Single-leg postural stability score with minimal center-of-mass drift.

4. Strength

Full-depth squat control and upper-body push-up rep volume.

Sport Weighting Matrix & Recommendation Model

Each sport maintains a distinct trait weighting vector and height calibration bonus:

Sport	Power	Agility	Balance	Strength	Coordination	Anthropometric Factor
Volleyball	**45%**	15%	5%	10%	25%	Height ≥ 180 cm bonus
Basketball	**40%**	25%	5%	10%	20%	Height ≥ 185 cm bonus
Football (Soccer)	20%	**35%**	10%	10%	25%	Balanced (170 - 182 cm)

Sport	Power	Agility	Balance	Strength	Coordination	Anthropometric Factor
Cricket	20%	25%	10%	10%	**35%**	Hand-eye / shoulder leverage
Badminton	25%	**40%**	5%	5%	25%	Agile, fast reaction speed
Tennis	15%	30%	10%	15%	**30%**	Rotational power / endurance

```
Suitability Score = (W_power * Power) + (W_agility * Agility) + (W_balance * Balance) +
(W_strength * Strength) + (W_coordination * Coordination) + HeightBonus
```

8. Comprehensive Talent Report & Offline Certificate

When all 5 assessments in the battery are completed, the system unlocks the **Comprehensive Talent Report** (`/talent-report`):

- **Definitive Sport Match:** Displays primary sport recommendation and alternative secondary sports.
- **4-Week Structured Practice Roadmap:** Formulates a progressive, week-by-week training regimen focused on the athlete's development areas.
- **Offline Official Certificate:** Generates a full printable certificate using client-side HTML/CSS rendering with athlete credentials, biometric verification seal, and talent score.

9. Interactive AI Biomechanics Coach Assistant

Accessible from any screen via the floating Coach icon (`AICoachDrawer.tsx`):

- **Olympic Benchmark Tiers:** Benchmarks performance against Olympic/Elite, State/College, Trained, and Beginner standards.
- **Dynamic Movement Tips:** Offers tailored kinematic cues (e.g. countermovement depth, eccentric loading, plyometric stiffness).
- **Interactive Chat:** Real-time conversational sports biomechanics guidance.

10. Database Schema & Local Storage Architecture

Supabase Cloud Database

• `profiles`: UUID, full_name, age, height, fitness_level, avatar_url, disclaimer_accepted.
• `user_sports`: User ID to sports catalog mapping.
• `assessment_runs`: Run ID, test slug, metric value, status, timestamp.

Client-Side Resilient State

• `sportsense.athlete_face_profile`: Vector ratios, photo snapshot, locked state.
• `sportsense.profile`: Athlete demographics & completeness.
• `TanStack Query v5`: Real-time query caching and invalidation across routes.

11. Local Setup & Build Pipeline

```
# 1. Clone & Install git clone https://github.com/Adarshhegde024/Sports_talent_assessment.git cd Sports_talent_assessment npm install # 2. Configure Environment (.env) VITE_SUPABASE_URL=https://your-supabase-project.supabase.co VITE_SUPABASE_ANON_KEY=your-supabase-anon-key # 3. Start Development Server npm run dev # http://localhost:5173 # 4. Production Build Verification npm run build # Generates optimized SSR & Client bundle in .vercel/output
```

Document Summary

SportSense AI Pro bridges computer vision, edge biometrics, and sports science to deliver a secure, hardware-free talent assessment standard for the next generation of athletes.